IoT Project: Performance Evaluation of LoRa Modulations for NTN-6G Satellite IoT

The objective of this project is to study and evaluate the performance of one of the main LoRa modulation schemes used for IoT communications, namely **LoRa CSS** or **LR-FHSS**, in a satellite communication scenario.

The project will be carried out in several steps:

1. Bibliographic study
Conduct an in-depth study of existing LoRa/LoRaWAN and Satellite-IoT simulators, based on the **survey paper by Khaoula** and the simulators listed in the provided table(Table I). The study should identify the main features, supported communication models, simulation environments, and parameters of each simulator.

2. Simulator selection and installation
Based on the bibliographic study, select a suitable open-source simulator that can be installed and used for the project. The students should justify their choice according to the objectives of the project and the features offered by the simulator

LoRa and LR-FHSS Simulators (Table I)

Simulator	Main Environment	Main Interest	Available Code
LoRaSim	Python / SimPy	LoRa collisions, ALOHA, capture effect, and scalability studies	https://github.com/mcbor/lorasim
FLoRa	OMNeT++ / INET	LoRaWAN network simulation, including nodes, gateways, Network Server, and ADR	https://github.com/florasim/flora
ns-3 LoRaWAN	ns-3 / C++	Detailed LoRaWAN modeling and network simulation	https://github.com/signetlabdei/lorawan
FLoRaSat	OMNeT++	Satellite extension of FLoRa, including satellite geometry/orbits and Satellite-IoT network simulation	https://gitlab.inria.fr/jfraire/florasat
LR-FHSS-Sim	Python	LR-FHSS simulation with frequency hopping and fragment-level modeling	https://github.com/Xexell/LR-FHSS-sim

3. Understanding and testing the simulator
Install and run the selected simulator using its default configuration in order to understand its architecture, inputs, outputs, and main simulation parameters. Initial experiments should be performed to verify that the simulator works correctly.

4. Performance evaluation
Modify the simulation parameters and study their impact on the communication performance. Depending on the capabilities of the selected simulator, parameters may include:

- satellite altitude;
- satellite speed or orbital parameters;
- number of IoT devices;
- traffic/load generated by IoT devices;
- transmission power;
- spreading factor or other modulation parameters;
- channel conditions and propagation parameters.

Relevant performance metrics may include **Packet Success Rate (PSR), Packet Delivery Ratio (PDR), collision rate, latency, throughput, and outage/access time**.

5. Analysis and comparison
Perform a set of simulation experiments and analyze how the selected parameters affect the performance of **CSS or LR-FHSS**. The results should be presented using appropriate graphs and tables, with a clear discussion of the observed behavior.

20 Slides presentation, withe state of the art and simulation results

Expected outcome:
The project should provide a reproducible simulation framework and a set of experiments allowing the students to understand the behavior and performance of a LoRa modulation scheme in a **Satellite-IoT environment**, while developing practical experience with simulation tools, parameter analysis, and scientific experimentation.